



**UNIVERSIDAD DE SAN CARLOS DE GUATEMALA
CENTRO UNIVERSITARIO DE OCCIDENTE
DEPARTAMENTO DE ESTUDIOS DE POSTGRADOS
“MAESTRIA EN INGENIERIA ESTRUCTURAL Y
SISMORRESISTENTE ”
DINAMICA ESTRUCTURAL APLICADA AL DISEÑO SISMICO
ING. CARLOS JULIÁN HERNÁNDEZ GARCÍA**

**“TAREA 2 ”
“EJEMPLOS MAPLE12”**

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FECHA

24 DE JUNIO 2026

PRIMER EJEMPLO

> *edo* := $y'' + \omega^2 y = 0$;

$$edo := \frac{d^2}{dx^2} y(x) + \omega^2 y(x) = 0 \quad (1)$$

> *cond* := $y(0) = u, D(y)(0) = 0$;

$$cond := y(0) = u, D(y)(0) = 0 \quad (2)$$

> *sol* := *dsolve*({*edo*, *cond*}, *y*(*x*));

$$sol := y(x) = u \cos(\omega x) \quad (3)$$

> *des* := *rhs*(*sol*);

$$des := u \cos(\omega x) \quad (4)$$

> *vel* := *diff*(*des*, *x*);

$$vel := -u \sin(\omega x) \omega \quad (5)$$

> *acel* := *diff*(*des*, *x*\$2);

$$acel := -u \cos(\omega x) \omega^2 \quad (6)$$

> *restart*;

>

> restart; **EJEMPLO 2**

> edo := y'' + 0.447²*y = 0;

$$edo := \frac{d^2}{dx^2} y(x) + 0.199809 y(x) = 0 \quad (1)$$

> cond := y(0) = 1, D(y)(0) = 2;

$$cond := y(0) = 1, D(y)(0) = 2 \quad (2)$$

> sol := dsolve({edo, cond}, y(x));

$$sol := y(x) = \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right) \quad (3)$$

> des := rhs(sol);

$$des := \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right) \quad (4)$$

> vel := diff(des, x);

$$vel := 2 \cos\left(\frac{447}{1000} x\right) - \frac{447}{1000} \sin\left(\frac{447}{1000} x\right) \quad (5)$$

> acel := diff(des, x\$2);

$$acel := -\frac{447}{500} \sin\left(\frac{447}{1000} x\right) - \frac{199809}{1000000} \cos\left(\frac{447}{1000} x\right) \quad (6)$$

> ur := 2.4504;

$$ur := 2.4504 \quad (7)$$

> vmax := 2.05;

$$vmax := 2.05 \quad (8)$$

> amax := 0.916;

$$amax := 0.916 \quad (9)$$

> plot([des, vel, acel, ur, vmax, amax], x = 0..20, y = -5..5, color = [blue, red, green, orange, orange, orange], legend = ["u(t)", "u'(t)", "u''(t)", "ur", "vmax", "amax"], title = "Vibración libre sin amortiguamiento", gridlines = true);

Vibración libre sin amortiguamiento

